Free body diagram and static finite element analysis of the human tibia

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Abstract: Static analysis on tibia is performed for three different postures of humans (standing, staircase ascent and descent) by drawing a free body diagram. A model of the tibia bone is created in CATIA software and is imported to ANSYS. Obtained values during standing and stair ascent and descent are incorporated in finite element analysis. The stress distribution and deformation made in the human bone tibia are observed.

Keywords: finite element analysis; nodes; meshing; tetrahedron; axial compressive stress; deformation; tibia.

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1 Introduction

In developing models such as a model of the tibia bone, the first step is to generate computer models of the bones. The finite element formulations of contact characteristics have been used by biomechanics investigators to evaluate relative shift of the bone. The elasticity of the bone is likely to be a factor influencing bone growth in the tibia. Static analysis was performed and material properties of bone were assumed linear, isotropic and inhomogeneous. Characteristics of the bone can be found underwater as well (Zhang et al., 2014). The rate of application of finite element analysis to the evaluation of stresses in bones and to the fracture fixation devices has increased drastically in the last decade. Evaluation of stresses in bones helps in defining the effects of the constituent properties of the bone on its macroscopic behaviour. The application of the finite element technique to orthopaedic problems is different from its application to structural problems in that, in orthopaedics, a lot is dependent on the nature of the individual living tissues involved. Material properties are not as predominantly defined as in structural problems. Both geometric characteristics and material properties play a very important role in results that are reached. Moreover, it is absolutely necessary that a multidisciplinary team (both engineers and doctors) be involved in the study of orthopaedic.

In recent days, replacement with implants in the joints of the human body is taking place. Materials used for implants are not much closer to the bone material (Baldwin et al., 2012; Halloran et al., 2005; Burny et al., 2000). Mechanical analysis should be done so that the exact forces may be determined, and depending upon the forces involved, suitable materials can be used. Owing to excessive loads transmitted to the bone, the ligaments may fail and may be exposed to severe pain (Meyer and Haut, 2005; Woo et al., 2006; Li et al., 1998). Implants may be designed in such a way that it reduces the muscle forces. With the mechanical analysis, muscle forces can also be obtained. Earlier authors used various optimisation techniques to reduce muscle forces (Pedotti et al., 1978; Challis and Kerwin, 1993). In the present paper, the analysis is done by reverse engineering the muscle, and axial compressive forces are obtained later the axial compressive loads are applied on the tibia bone in various postures in ANSYS; thereafter, stress distribution are found. Here due to abduction and adduction movement of the knee, it is difficult to apply forces in the tibia bone; these loads are of varying loads (Harrington, 1983).

Computer Tomography (CT) scans or Magnetic Resonance Imaging (MRI) constitutes the input data required for modelling the human parts such as bones. Using these data we can make FEM model (Pena et al., 2006; Devika and Arumaikkannu, 2011) incorporating the material properties and the loading conditions, which yield required results. Since bone is an anisotropic material the properties in three directions must be used. In the present work, we got the dimensions through X-ray and MRI scans. We used the anisotropic properties of the tibia bone. The model is first created in CATIA software;

later it is imported to ANSYS software here meshing is done by using a ten-node solid tetrahedron with 4723 elements. Forces acting on tibia while standing, staircase ascent and staircase descent (Madeti et al., in press) considering the forces on the knee stress distribution can be clearly defined by finite element analysis (Madeti et al., 2014a, Tsouknidas et al., 2012). In contrast, while considering the transmission of loads to the knee, in the case of descent, the load acting on the knee is nearly 3.2–4.9 times the body weight. In the case of ascent, the load is 2.8–4.4 times the body weight (Komistek et al., 2005).

2 Forces acting on knee during ascending and descending staircase

During staircase ascent and descent, forces involved are muscle force and axial compressive force which are the unknown forces. Applying equilibrium conditions, for various positions, values have been determined as shown in Figure 1:

- $m_1 g$ = weight of leg;
- R = normal reaction from staircase;
- N = axial compressive force on the tibia bone;
- F_q = muscle force;
- α = angle made by N with the vertical axis;
- θ = angle made by F_q with the horizontal axis;

Resolving forces along x-axis,

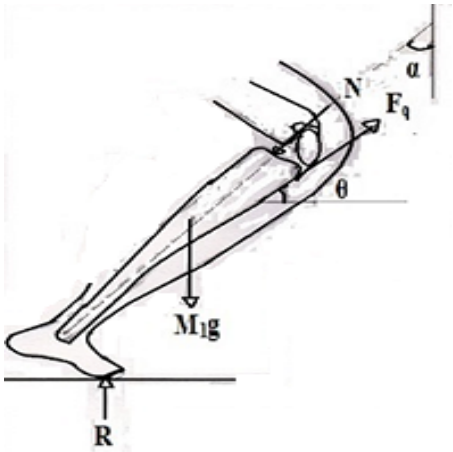
$$\sum F_x = 0$$
$$F_q \cos\theta = N \sin\alpha \tag{1}$$

Resolving forces along x-axis,

$$\sum F_y = 0$$
$$F_q \sin\theta + R = m_1 g + N \cos\alpha \tag{2}$$

The above equations are obtained from Figure 1. By solving these equations, forces acting on tibia are attained. Consider a male of 60 kg body mass. Then, m_1 = 2.1 kg, R = 60 × 9.81 and we also found that α = 40° and the corresponding θ = 40° by substituting these values in the above equations. Consider a male with body mass 60 kg, so m_1 = 2.1 kg; R = 60 * 9.81; we found that α = 40° and the corresponding θ = 220° by substituting these values in the above equations. The axial compressive force on tibia is 1800 N force acting on the knee during staircase ascent, while descending forces acting on the knee is 2500 N; in the standing posture, half the weight of the person is considered (Madeti et al., 2014b). Different combinations of various forces may obtained by taking the best values in the finite element analysis.

Figure 1 Various forces acting on knee



Source: Halloran et al. (2005)

3 Process of creating FE model of the tibia

The process of creating an FE model of the tibia is as follows:

1 The tibia of a 27-year-old male weighing 60 kg with tibia length of 30 cm and the area of the top of the tibia of 2795 mm^2 is shown in Figure 2.

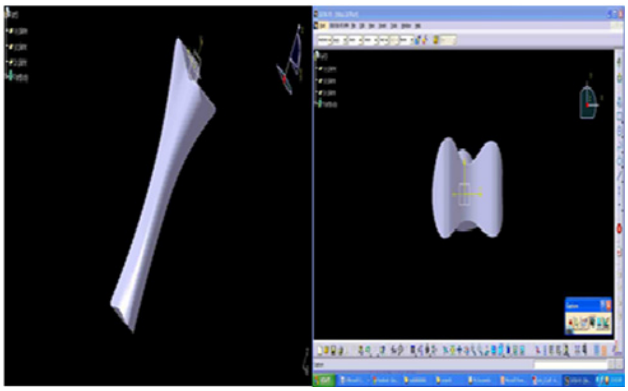
2 Multi-section solid command is used to get different cross-sections.

3 The fibula bone is a supporting bone to the tibia; the forces acting on it is neglected, since load transmitted to the fibula bone is neglected.

4 Later, cut the top of the tibia with a surface having the same profile as that of the tibial surface.

5 Later, it is saved in the iges.file format.

Figure 2 Different views of tibia model in CATIA



4 FE model for analysis of tibia

1 The CATIA file saved in iges.file is imported in ANSYS Mechanical APDL software.

2 Later the material properties were given, such as the young's modulus, shear modulus and Poisson ratio in all three directions in Tables 1 and 2.

3 We used a ten-node solid tetrahedron as the mesh element.

4 The meshing was done in fine mesh mode as shown in Figure 3.

5 Boundary conditions are applied as shown in Figure 4:

- Lower end of the tibia was fixed in all three DOF where a uniform pressure of 210 KPa is applied, i.e. the maximum pressure while standing was applied on the top of the tibia. The stress distribution and deformation are shown in Figures 5 and 6.
- In a similar boundary condition, a uniform pressure of 421 KPa was applied simulating the condition of a staircase ascent; the stress distribution and deformation are shown in Figures 7 and 8.
- A 631 KPa uniform pressure simulates the condition of descending a staircase; the stress distribution and deformation is shown in Figures 9 and 10.

6 The element used is a ten-node solid tetrahedron with 4723 elements.

As shown in Figure 11, a comparison of the maximum stresses in the human tibia bone while standing, staircase ascent and staircase descent is shown clearly.

Table 1 Material properties of tibia in all three directions

Material property	*Value in GPa*
E1	4.48
E2	4.48
E3	9.64
G12	1.41
G13	1.28
G23	1.28

Table 2 Poisson ratio in all three directions

Poisson ratio	*Value*
υ_{12}	0.35
υ_{13}	0.12
υ_{23}	0.12

Figure 3 Meshed tibia

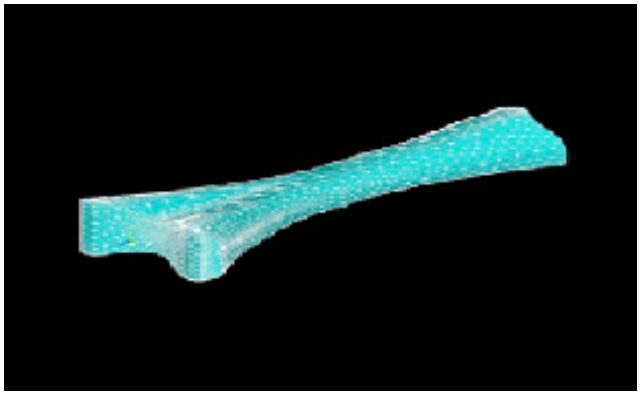


Figure 4 Tibia with boundary conditions

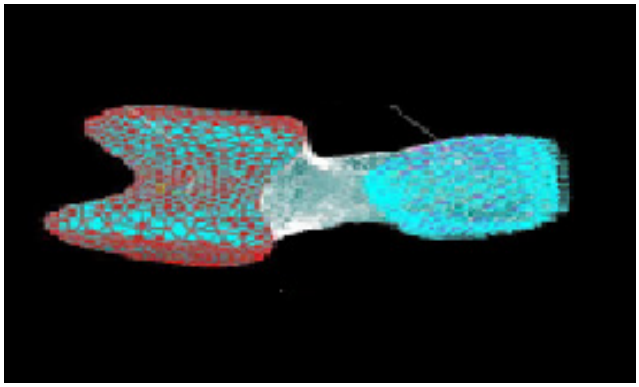


Figure 5 Deformed shape while standing

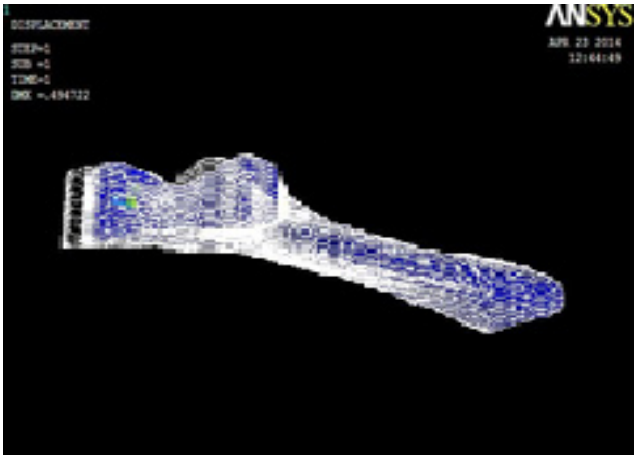


Figure 6 Stress distribution while standing

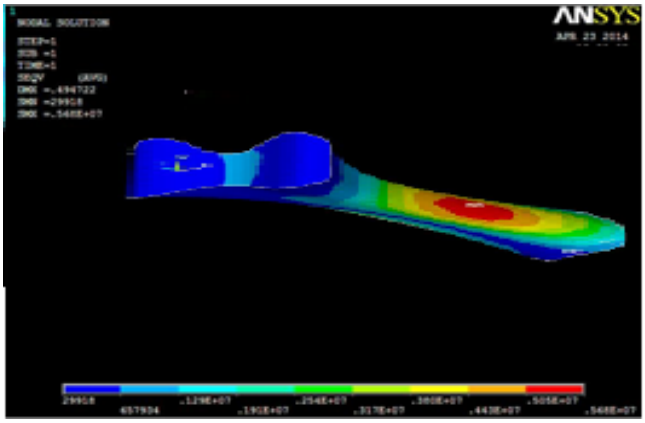


Figure 7 Deformed shape in stair ascent

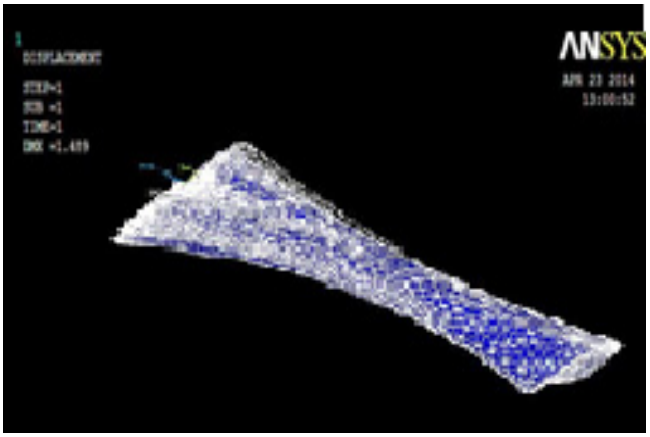
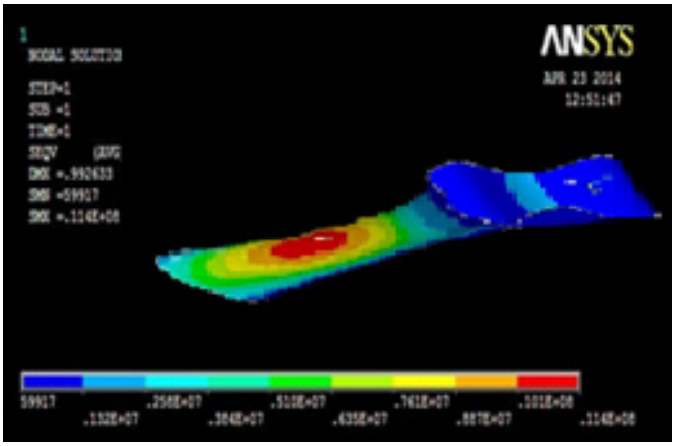


Figure 8 Stress distribution in stair descent



As shown in Figure 12, several nodal points are taken on the surface of the tibia; the graph is plotted between the three cases of standing position, stair ascent and stair descent owing to this compressive force which acts along the human tibia. Here comparisons of stresses in three cases are shown clearly.

Figure 9 Deformed shape in stair descent

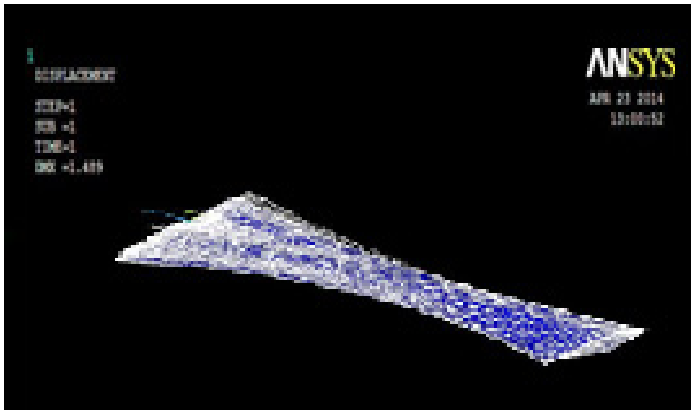


Figure 10 Stress distribution in stair descent

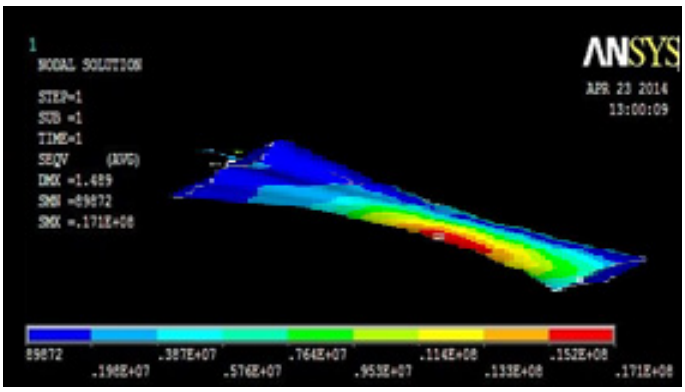


Figure 11 Axial compressive force on tibia (MPa)

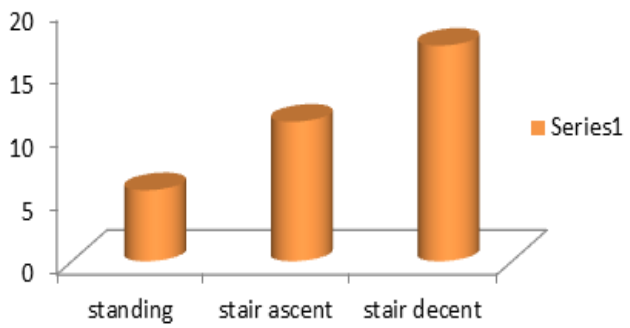
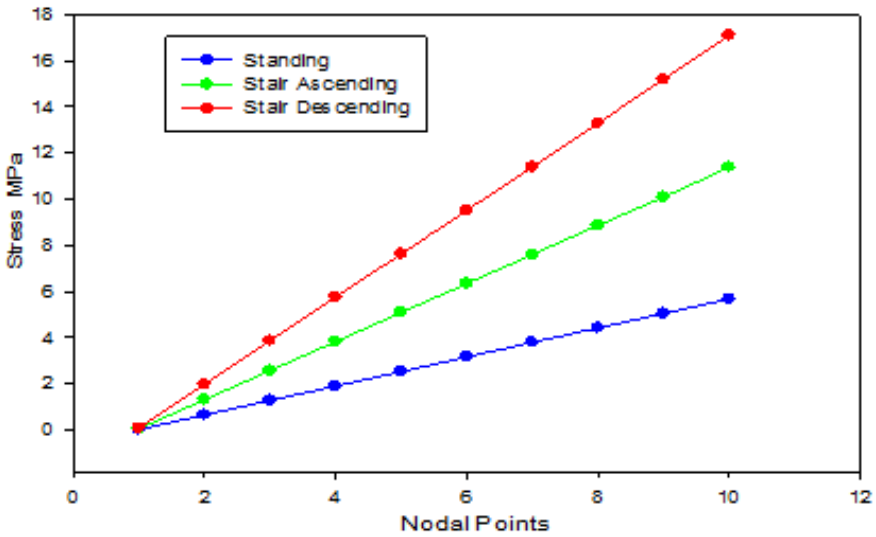


Figure 12 Stresses obtained at several nodal points at the surface of tibia



5 Results and discussion

Results obtained from the finite element analysis are the maximum compressive stress on the tibia during the three postures of standing, staircase ascent and staircase descent: 5.6, 11 and 17 MPa, respectively. The maximum deformation occurring in the three cases are 0.49, 1.489 and 2.004 mm, respectively, for 600*N* person. For all these cases, more stresses are developed in the lower portion of the tibia. We have considered Von Mises effective stress because the bone is a ductile material. The Tresca stress is usually used for brittle materials.

6 Conclusions

The conclusions of the study are as follows:

- The axial compressive force and muscle force can be obtained by drawing free body diagram of the human knee.
- During stair ascent, less axial compressive force is required, while descending the axial compressive is more.
- An FEA analysis of the tibia with this stress distribution is clearly identified throughout the surface of the tibia.
- By knowing the maximum stresses, suitable replacement components may be designed. Materials can be selected, which are close to the material properties of the bone.

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